Autoregressive Processes, Delay Embeddings, and TDA-Based Scale Selection for Estimating the Order p

A Detailed Report for Time-Series Order Estimation

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Abstract

This report explains autoregressive processes of order p, abbreviated AR(p), and develops a practical framework for estimating p using a topological data analysis (TDA) pipeline based on delay embeddings, Vietoris–Rips complexes, and spectral scale selection. The central idea is that a scalar time series can be converted into a sequence of point clouds by sliding-window or delay-coordinate embeddings. For an AR(p) process, the conditional law of the next observation depends on the last p observations; therefore, embedding dimensions below p miss relevant dependence, whereas dimensions above p add increasingly redundant coordinates. TDA can be used to detect this transition by tracking how multiscale simplicial-complex summaries, Betti numbers, f-vectors, LS-category-inspired point-complexity bounds, and spectral critical scales stabilize as the embedding dimension grows.

The report also discusses Takens' embedding theorem. Takens' theorem is a deterministic theorem about reconstructing compact attractors from generic observations; an AR(p) process is stochastic and therefore does not satisfy Takens' hypotheses in a literal sense. Nevertheless, Takens-type delay coordinates provide a useful conceptual and computational language for embedding a scalar time series into Euclidean space. The proposed method should therefore be viewed not as a direct theorem proving that p is recovered exactly, but as a statistically motivated diagnostic procedure: choose p as the embedding dimension at which the topological and geometric information added by further delays becomes small or structurally redundant.

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1 Motivation and High-Level Idea

Classical autoregressive order estimation is usually based on autocorrelation functions, partial autocorrelation functions, information criteria such as AIC or BIC, likelihood-based model selection, or prediction error. These methods are powerful because an AR(p) model has a highly explicit probabilistic structure. However, they are mostly linear-statistical tools. If the observed data are noisy, non-Gaussian, heavy-tailed, locally nonlinear, contaminated by outliers, or generated by a process that is only approximately autoregressive, it can be useful to supplement classical order estimation with geometric and topological diagnostics.

The TDA approach developed here starts from a simple principle:

> Instead of looking only at scalar correlations, convert the time series into point clouds of increasing delay dimension and ask when the shape of these point clouds stops changing in an essential way.

For a scalar time series $(X_t)_{t\in\mathbb{Z}}$, define for each embedding dimension k a delay vector

$$Y_t^{(k)} = (X_t, X_{t-\tau}, X_{t-2\tau}, \ldots, X_{t-(k-1)\tau}) \in \mathbb{R}^k, \tag{1}$$

where $\tau \geq 1$ is the delay. For simplicity, most of this report uses $\tau = 1$ unless otherwise stated. The embedded point cloud is

$$\mathcal{Y}_k = \{Y_t^{(k)} : t = (k-1)\tau + 1, \ldots, n\} \subseteq \mathbb{R}^k. \tag{2}$$

For every k, one builds a filtration of Vietoris–Rips complexes

$$\mathrm{VR}_\varepsilon(\mathcal{Y}_k), \qquad \varepsilon \in \mathcal{E}_k, \tag{3}$$

extracts topological and combinatorial summaries, and selects an informative scale $\varepsilon_{\mathrm{crit}}(k)$ using a spectral procedure. The order estimate is then obtained by detecting a transition across embedding dimensions k.

1.1 Why this might estimate an AR order

An AR(p) process satisfies

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \tag{4}$$

where (Z_t) is an innovation sequence. Thus, once $(X_{t-1}, \ldots, X_{t-p})$ is known, older values do not add further linear predictive information in the ideal model. In Markov language, the scalar process becomes first-order Markov after lifting it to the p-dimensional state vector

$$S_t = (X_t, X_{t-1}, \ldots, X_{t-p+1}) \in \mathbb{R}^p. \tag{5}$$

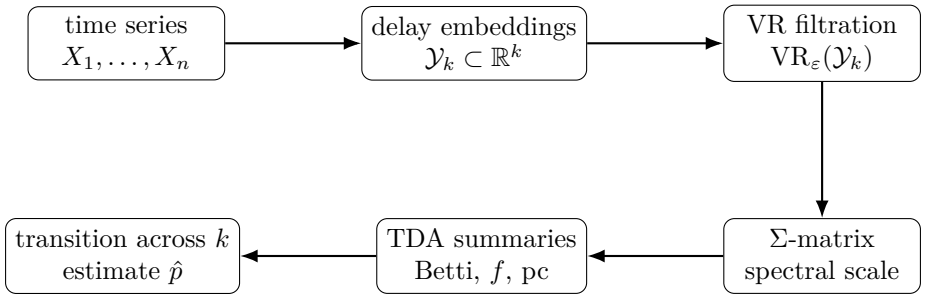
Therefore, a sliding-window embedding of dimension $k < p$ is missing part of the state, while $k \geq p$ contains the full autoregressive state. This gives a natural topological order-selection heuristic:

> Increase k and monitor how the topology, combinatorics, and selected scale of $\mathcal{Y}_k$ change. Estimate p near the first dimension where the summaries stabilize or where a clear transition occurs.

1.2 Relation to the existing scale-selection pipeline

The pipeline used in the accompanying internship work constructs Vietoris–Rips complexes across a grid of ε values, computes LS-category-inspired characteristics, forms a similarity or Σ-matrix, and uses the principal eigenvector to identify a critical scale. In that setting, the scale $\varepsilon_{\text{crit}}$ is intended to reveal the scale at which the relevant topological structure of a point cloud is most visible. Here the same philosophy is applied repeatedly, once for each delay dimension k.

The conceptual extension is as follows:



2 Autoregressive Processes of Order p

2.1 Definition

A real-valued time series (X_t) is called an autoregressive process of order p, or AR(p), if it satisfies

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \tag{6}$$

where $c \in \mathbb{R}$ is an intercept, $\phi_1, \ldots, \phi_p \in \mathbb{R}$ are autoregressive coefficients, and (Z_t) is a white-noise innovation sequence, commonly assumed to satisfy

$$\mathbb{E}[Z_t] = 0, \qquad \mathrm{Var}(Z_t) = \sigma_Z^2, \qquad \mathrm{Cov}(Z_t, Z_s) = 0 \quad (t \neq s). \tag{7}$$

If the mean is $\mu = \mathbb{E}[X_t]$, then under stationarity

$$\mu = \frac{c}{1 - \phi_1 - \cdots - \phi_p}, \tag{8}$$

provided the denominator is nonzero. Often one centers the data and works with $\tilde{X}_t = X_t - \mu$, giving

$$\tilde{X}_t = \phi_1 \tilde{X}_{t-1} + \cdots + \phi_p \tilde{X}_{t-p} + Z_t. \tag{9}$$

2.2 Backshift notation and characteristic polynomial

Let B be the backshift operator, $BX_t = X_{t-1}$. The AR(p) equation can be written compactly as

$$\phi(B) X_t = c + Z_t, \tag{10}$$

where

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p. \tag{11}$$

A standard sufficient and necessary condition for causal weak stationarity is that all roots of $\phi(z) = 0$ lie outside the unit circle. Under this condition the process admits an infinite moving-average representation

$$X_t - \mu = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \tag{12}$$

with absolutely summable coefficients (ψ_j).

2.3 AR(p) as a p-dimensional Markov state model

The scalar AR(p) recursion can be rewritten as a first-order vector recursion. Define

$$S_t = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-p+1} \end{pmatrix} \in \mathbb{R}^p, \qquad \eta_t = \begin{pmatrix} Z_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{13}$$

Then

$$S_t = AS_{t-1} + b + \eta_t, \tag{14}$$

where, for the centered case,

$$A = \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}. \tag{15}$$

This representation is crucial for the TDA approach: the natural state space of an AR(p) process is p-dimensional. A delay embedding of dimension k attempts to reconstruct or approximate this state space from observations.

2.4 Classical order estimation

Classical methods estimate p using model fit, autocorrelation, or prediction. The most common tools are:

(i) **Autocorrelation function (ACF).** The ACF of an AR(p) process usually tails off rather than cutting off.

(ii) **Partial autocorrelation function (PACF).** In the ideal AR(p) case, the PACF cuts off after lag p.

(iii) **AIC/BIC.** Fit AR(q) models for $q = 0, \ldots, q_{\max}$ and choose the value minimizing an information criterion.

(iv) **Prediction error.** Split the series into training and validation blocks and choose the model order that gives good out-of-sample prediction.

These tools should remain part of the empirical analysis. A TDA-based method is best viewed as a complementary diagnostic, especially when one wants to study geometry, nonlinear structure, outliers, or robustness.

3 Delay Embedding of a Time Series

3.1 Sliding-window point clouds

Given observations $x_1, \ldots, x_n$, an embedding dimension k, and a delay τ, define

$$y_t^{(k,\tau)} = (x_t, x_{t-\tau}, \ldots, x_{t-(k-1)\tau}) \in \mathbb{R}^k. \tag{16}$$

The corresponding point cloud is

$$\mathcal{Y}_{k,\tau} = \{y_t^{(k,\tau)} : t = (k-1)\tau + 1, \ldots, n\}. \tag{17}$$

For $\tau = 1$ we write simply $\mathcal{Y}_k$.

3.2 Preprocessing

Preprocessing matters because TDA is sensitive to distances. A recommended pipeline is:

(1) Remove or estimate the mean. For AR modeling, use centered values unless the intercept is explicitly modeled.

(2) Optionally detrend if the series contains a deterministic trend. AR order estimation assumes at least approximate stationarity.

(3) Standardize the series to unit variance before building embeddings.

(4) For each k, optionally whiten or rescale coordinates if different lags have strongly different empirical variances. For stationary AR processes, lag coordinates should have similar marginal variance, so simple standardization of the original series is usually enough.

(5) Choose a maximum embedding dimension $k_{\max}$. A typical range is $k = 1, \ldots, 10$ or $k = 1, \ldots, 20$, depending on sample size.

3.3 Choice of delay τ

For AR(p) order estimation, the default choice is $\tau = 1$, because the model itself uses consecutive lags. Choosing $\tau > 1$ can be useful for strongly oversampled continuous-time signals, but it changes the interpretation: the embedding contains $X_t, X_{t-\tau}, X_{t-2\tau}, \ldots$ rather than the consecutive state. Therefore, if the goal is to estimate the discrete AR order p, use $\tau = 1$ as the main analysis and treat $\tau > 1$ as a sensitivity check.

4 Takens' Theorem and Its Role

4.1 The deterministic theorem

Takens' embedding theorem is one of the conceptual foundations of time-delay reconstruction. In a simplified form, suppose M is a compact smooth manifold of dimension d, $F : M \to M$ is a generic smooth diffeomorphism, and $h : M \to \mathbb{R}$ is a generic smooth observation function. The delay-coordinate map

$$\Phi_{h,F}(x) = \big(h(x), h(F(x)), \ldots, h(F^m(x))\big) \tag{18}$$

embeds M into $\mathbb{R}^{m+1}$ for $m \geq 2d$. Equivalently, $m+1 \geq 2d+1$ scalar observations are generically enough to reconstruct the attractor without self-intersections.

4.2 Why Takens does not directly prove AR order recovery

An AR(p) process is stochastic. The next value is not determined by the previous state alone; it also depends on the innovation Z_t. Therefore, the observed sequence is not generated by iterating a deterministic diffeomorphism on a compact manifold in the way required by Takens' theorem. Consequently:

- Takens' theorem should not be cited as a direct proof that delay embedding recovers the AR order p.
- The theorem does justify the broader idea that time-delay vectors can reveal hidden state-space geometry.
- For stochastic processes, embedding is better understood through Markov state reconstruction, conditional distributions, and statistical dependence rather than deterministic attractor reconstruction.

4.3 How Takens still helps conceptually

Takens' theorem motivates the method in three ways. First, it says that scalar observations can contain enough information to reconstruct a state space. Second, it suggests that the embedding dimension is a structural parameter: too small an embedding can create artificial self-overlaps, whereas a sufficiently large embedding can unfold the dynamics. Third, it encourages studying the geometry of the embedded point cloud rather than only scalar summaries.

For an AR(p) process, the relevant "state" is not a deterministic attractor but the p-lag vector $(X_t, \dots, X_{t-p+1})$. Therefore the TDA version of the Takens intuition is:

> The point cloud $\mathcal{Y}_k$ should become a sufficiently informative state-space representation once k reaches the memory length p.

5 Geometric Structure of AR(p) Delay Clouds

5.1 Noise-free intuition

Suppose first that $Z_t = 0$ and $c = 0$. Then

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p}. \tag{19}$$

In the $(p+1)$-dimensional delay vector

$$(X_t, X_{t-1}, \dots, X_{t-p}) \in \mathbb{R}^{p+1}, \tag{20}$$

the coordinates satisfy one linear constraint:

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = 0. \tag{21}$$

Thus the noiseless $(p+1)$-delay cloud lies in a p-dimensional hyperplane. More generally, for $k > p$, the delay coordinates satisfy linear recurrence relations, creating a low-dimensional structured set inside $\mathbb{R}^k$.

5.2 Stochastic AR(p) intuition

When $Z_t \neq 0$, the hyperplane constraint becomes

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = Z_t. \tag{22}$$

The embedded cloud is no longer exactly constrained to a hyperplane; it becomes a noisy tube or ellipsoidal cloud whose thickness depends on the innovation variance and the coefficient structure. Still, the process has memory p: older coordinates affect X_t only through their correlation with the last p coordinates. Hence, increasing k beyond p should add redundant geometry rather than fundamentally new state information.

5.3 Expected changes as k increases

The following table summarizes the qualitative expectation.

Embedding dimension	Statistical meaning	Expected TDA behavior

$k < p$	The delay vector misses part of the AR state.	Point cloud may be projected, folded, or geometrically compressed; topological summaries may change strongly with k.
$k = p$	The delay vector contains the Markov state.	Important transition point; topology and scale summaries may become more stable.
$k = p + 1$	Adds one coordinate governed by the AR relation plus innovation.	May reveal a constraint/tube structure; can be especially diagnostic through scale, simplex counts, and local dimension.
$k > p + 1$	Adds increasingly redundant older lags.	Summaries should vary more slowly; changes mainly reflect sampling, noise, and high-dimensional distance effects.

This is not a theorem guaranteeing exact recovery, but it is a precise experimental hypothesis.

6 Vietoris–Rips Complexes and Persistent Homology

6.1 Vietoris–Rips complex

Let $X = \{x_1, \ldots, x_N\} \subseteq \mathbb{R}^d$ be a finite metric point cloud. The Vietoris–Rips complex at scale ε is

$$\mathrm{VR}_\varepsilon(X) = \{\sigma \subseteq X : \|x_i - x_j\| \leq \varepsilon \text{ for all } x_i, x_j \in \sigma\}. \tag{23}$$

Equivalently, a simplex is included whenever all its vertices are pairwise within distance ε. As ε increases, the complexes are nested:

$$\varepsilon_1 \leq \varepsilon_2 \quad \Rightarrow \quad \mathrm{VR}_{\varepsilon_1}(X) \subseteq \mathrm{VR}_{\varepsilon_2}(X). \tag{24}$$

At very small ε, the complex consists mostly of isolated vertices. At very large ε, the complex approaches a full simplex and becomes contractible. Interesting topology usually appears at intermediate scales.

6.2 Betti numbers

The qth Betti number β_q counts independent q-dimensional holes:

- β_0: connected components,
- β_1: loops or one-dimensional cycles,
- β_2: voids or two-dimensional cavities.

For time-series delay clouds, one should not expect simple textbook shapes such as perfect circles or tori in general. Instead, the useful information is often contained in how Betti numbers change across k and ε.

6.3 f-vectors

The f-vector of a simplicial complex records the number of simplices by dimension:

$$f(K) = (f_0(K), f_1(K), f_2(K), \ldots), \tag{25}$$

where $f_q(K)$ is the number of q-simplices. For a Vietoris–Rips complex, $f_0 = N$ is the number of points, f_1 is the number of edges, and f_2 is the number of filled triangles. The f-vector is not a topological invariant, but it is highly informative about density, local connectivity, and scale.

6.4 Why scale selection is necessary

A single point cloud does not have a single canonical Vietoris–Rips complex. The scale ε controls which distances are considered local. If ε is too small, the complex is disconnected and dominated by sampling noise. If ε is too large, all structure is filled in. Therefore, before comparing embedding dimensions k, one needs a principled way to choose an informative scale for each $\mathcal{Y}_k$.

7 Spectral Scale Selection Using an LS-Category-Inspired Matrix

7.1 LS-category and point-complexity proxy

The Lusternik–Schnirelmann category $\mathrm{cat}(K)$ measures, roughly, how many contractible pieces are needed to cover a space or complex. Directly computing LS-category can be difficult. The internship pipeline uses a computable proxy based on point complexity. For a vertex $x \in X$ and a complex $K = \mathrm{VR}_\varepsilon(X)$, define a local quantity of the form

$$pc_\varepsilon(x) = d - \max\{\dim(\sigma) : x \in \sigma,\, \sigma \in K\}, \tag{26}$$

where d is an ambient or reference dimension. Summing over vertices gives an upper-bound-style complexity score

$$PC(\mathrm{VR}_\varepsilon(X)) = \sum_{x \in X} pc_\varepsilon(x). \tag{27}$$

The exact implementation can include corrections using local edge and triangle coverage; the important point is that PC decreases or changes when vertices become well integrated into higher-dimensional simplices.

7.2 A scalar product between complexes

Given complexes $K_i = \mathrm{VR}_{\varepsilon_i}(X)$ and $K_j = \mathrm{VR}_{\varepsilon_j}(X)$, define a similarity score inspired by category inclusion-exclusion:

$$\langle K_i, K_j \rangle = -\mathrm{cat}(K_i \cup K_j) + \mathrm{cat}(K_i) + \mathrm{cat}(K_j). \tag{28}$$

In practice, cat is replaced by a computable proxy such as PC. This gives a matrix

$$\Sigma_{ij} = \langle \mathrm{VR}_{\varepsilon_i}(X), \mathrm{VR}_{\varepsilon_j}(X) \rangle. \tag{29}$$

The matrix summarizes how complexes at different scales relate to one another.

7.3 Critical scale from the principal eigenvector

Assume Σ is nonnegative or has been shifted/processed so that Perron–Frobenius intuition applies. Let v^* be the principal eigenvector corresponding to the largest eigenvalue. Define

$$\varepsilon_{\mathrm{crit}} = \varepsilon_{i^*}, \qquad i^* = \arg\max_i v_i^*. \tag{30}$$

The coordinate with largest eigenvector weight identifies the scale most central to the similarity structure of the filtration. This is the selected TDA scale.

7.4 Important implementation detail: eigenvector sign

If one computes eigenvectors numerically for a symmetric matrix, the sign is arbitrary: both v and $-v$ are valid eigenvectors. Therefore, if the principal eigenvector has all negative entries, one may multiply it by -1 before taking the largest coordinate. What matters is the ray spanned by the eigenvector, not the sign returned by the numerical routine. If the matrix is genuinely nonnegative and irreducible, the Perron vector can be chosen strictly positive, but software may still return its negative depending on conventions.

8 Estimating AR Order with the TDA Pipeline

8.1 Core algorithm

Let $x_1, \dots, x_n$ be a univariate time series. Choose $k_{\max}$ and a delay $\tau = 1$. For each embedding dimension $k = 1, \dots, k_{\max}$:

Step 1. Build the delay point cloud

$$\mathcal{Y}_k = \{(x_t, x_{t-1}, \dots, x_{t-k+1}) : t = k, \dots, n\} \subseteq \mathbb{R}^k. \tag{31}$$

Step 2. Standardize the coordinates if needed.

Step 3. Construct an ε grid $\mathcal{E}_k = \{\varepsilon_1, \dots, \varepsilon_M\}$, for example using quantiles of pairwise distances.

Step 4. For each ε_i, build $\mathrm{VR}_{\varepsilon_i}(\mathcal{Y}_k)$ up to a chosen maximum simplex dimension.

Step 5. Compute complexity scores PC_i, f-vectors, and Betti numbers.

Step 6. Build the $\Sigma^{(k)}$ matrix from the chosen similarity formula.

Step 7. Compute the principal eigenvector $v^{(k)}$ and select $\varepsilon_{\mathrm{crit}}(k)$.

Step 8. At $\varepsilon_{\mathrm{crit}}(k)$, record a feature vector

$$T(k) = (\varepsilon_{\mathrm{crit}}(k), PC(k), \beta_0(k), \beta_1(k), \beta_2(k), f_0(k), f_1(k), f_2(k), \dots). \tag{32}$$

Finally, inspect the sequence $T(1), \dots, T(k_{\max})$ and choose $\hat{p}$ as a transition or stabilization point.

8.2 Transition criteria

Several concrete criteria can be used. It is best to compute multiple criteria and look for agreement.

8.2.1 First stabilization of critical scale

Define

$$\Delta_\varepsilon(k) = \frac{|\varepsilon_{\mathrm{crit}}(k+1) - \varepsilon_{\mathrm{crit}}(k)|}{\varepsilon_{\mathrm{crit}}(k) + \delta}, \tag{33}$$

where $\delta > 0$ avoids division by zero. Estimate p as the first k after which $\Delta_\varepsilon(k)$ remains small for several consecutive dimensions.

8.2.2 Feature-vector increments

Let $T(k)$ be a normalized vector of TDA summaries. Define

$$D(k) = \|T(k+1) - T(k)\|. \tag{34}$$

A large drop in $D(k)$ suggests that additional delays stop changing the topology substantially. One can choose

$$\hat{p} = \arg\min_{k \geq k_{\min}} D(k) \tag{35}$$

or choose the first k for which $D(k)$ falls below a threshold.

8.2.3 Elbow in a cumulative curve

Define a scalar score $S(k)$, such as the selected-scale point complexity or a bilinear combination of Betti and f-vector entries. Then identify an elbow in $S(k)$ or in the cumulative variation

$$C(k) = \sum_{j=1}^{k-1} \|T(j+1) - T(j)\|. \tag{36}$$

The estimated order is the elbow where additional embedding dimensions provide diminishing returns.

8.2.4 Stability under bootstrap or block subsampling

For time series, ordinary independent bootstrap is inappropriate because it destroys dependence. Instead, use block subsampling. For each block-resampled series, compute $\hat{p}_b$. Then report the distribution of $\hat{p}_b$. A reliable estimate should be stable under moderate block resampling.

8.3 A practical decision rule

A practical rule combining the above ideas is:

$$\hat{p} = \min \left\{ k : \frac{1}{L} \sum_{\ell=0}^{L-1} \|T(k+\ell+1) - T(k+\ell)\| \leq \theta \right\}, \tag{37}$$

where L is a small persistence window such as 2 or 3, and θ is chosen from a null or surrogate distribution. The purpose of the persistence window is to avoid selecting a dimension due to one accidental small change.

9 Choice of Topological Features

9.1 Critical scale features

The sequence $k \mapsto \varepsilon_{\text{crit}}(k)$ is often one of the most useful outputs. However, comparing raw ε values across dimensions is delicate because distances in $\mathbb{R}^k$ grow with k. Therefore, one should use normalized versions, such as

$$\tilde{\varepsilon}_{\text{crit}}(k) = \frac{\varepsilon_{\text{crit}}(k)}{\text{median}\{\|y_i - y_j\| : i < j\}}, \tag{38}$$

or choose the ε grid by distance quantiles and compare the selected quantile index rather than the raw scale.

9.2 Betti features

At $\varepsilon_{\text{crit}}(k)$, record

$$B(k) = (\beta_0(k), \beta_1(k), \beta_2(k)). \tag{39}$$

For AR clouds, β_0 is related to connectivity and clustering. β_1 may indicate loops caused by oscillatory behavior, quasi-periodicity, or projected dependence structure. β_2 is harder to estimate reliably unless the sample size is large.

9.3 f-vector features

The f-vector

$$F(k) = (f_0(k), f_1(k), f_2(k), \ldots) \tag{40}$$

captures how many simplices appear at the selected scale. Since f_0 is mostly sample size, more informative normalized features are

$$\frac{f_1(k)}{\binom{N_k}{2}}, \qquad \frac{f_2(k)}{\binom{N_k}{3}}, \tag{41}$$

where $N_k = n - k + 1$ is the number of delay vectors. These are edge and triangle densities.

9.4 Point-complexity features

The LS-category-inspired point-complexity score is useful because it reflects how locally integrated points are in the simplicial complex. Outliers and isolated points contribute differently from points lying in high-dimensional local neighborhoods. For order estimation, useful features include:

- total point complexity $PC(k)$,
- mean point complexity $PC(k)/N_k$,
- number or proportion of isolated vertices at $\varepsilon_{\text{crit}}(k)$,
- corrected point complexity after removing isolated vertices.

9.5 Bilinear combinations

Suppose $b(k)$ is a vector of Betti summaries and $f(k)$ is a vector of simplex-count summaries. One can define a bilinear score

$$Q_A(k) = b(k)^\top A f(k), \tag{42}$$

where A is a chosen matrix. This generalizes simple products such as "number of connected components times number of edges." A systematic strategy is to start with interpretable matrices A and then evaluate which scores best separate known AR orders in simulation.

10 Simulation Design for Validating the Method

10.1 Synthetic AR(p) experiments

To test the method, generate AR processes with known order:

$$\text{AR(1):} \quad X_t = 0.7X_{t-1} + Z_t, \tag{43}$$

$$\text{AR(2):} \quad X_t = 1.2X_{t-1} - 0.5X_{t-2} + Z_t, \tag{44}$$

$$\text{AR(3):} \quad X_t = 0.6X_{t-1} + 0.2X_{t-2} - 0.3X_{t-3} + Z_t. \tag{45}$$

Use several innovation distributions:

- Gaussian innovations,
- Student-t innovations for heavy tails,
- contaminated innovations with occasional large outliers,
- heteroskedastic innovations as a robustness stress test.

10.2 Expected results

For each true order p, plot the following against embedding dimension k:

- selected scale index or normalized $\varepsilon_{\text{crit}}(k)$,
- $PC(k)$ and corrected $PC(k)$,
- edge density and triangle density,
- Betti vector at critical scale,
- norm of feature increments $D(k)$,
- final estimated order $\hat{p}$.

A successful method should show a visible transition near $k = p$ or $k = p + 1$. Which of these appears depends on the chosen summary. If the summary is based on the predictive state, the transition may occur at p. If it is based on detecting the recurrence relation in $(X_t, \ldots, X_{t-p})$, the strongest signal may occur at $p + 1$ because the AR equation first becomes directly visible as a constraint involving $p + 1$ consecutive values.

10.3 Comparison with classical estimators

The TDA estimate should be compared with:

- PACF cutoff,
- AIC-selected AR order,
- BIC-selected AR order,
- cross-validated one-step prediction error.

The most convincing result is not necessarily that TDA always beats classical methods, but that it provides additional diagnostic information. For example, if PACF suggests $p = 2$ but TDA shows no stabilization until $k = 4$, this may indicate nonlinear dependence, regime switching, unmodeled seasonality, or outliers.

11 Implementation Pseudocode

11.1 Python-style pseudocode

Listing 1: High-level pseudocode for TDA-based AR order estimation.

```
def estimate_ar_order_tda(x, k_max=12, tau=1, n_eps=40, max_dim=2):
    # 1. Preprocess
    x = np.asarray(x, dtype=float)
    x = x - np.mean(x)
    x = x / np.std(x)
```

```
    all_features = []

    for k in range(1, k_max + 1):
        # 2. Delay embedding
        Y = delay_embed(x, dim=k, tau=tau)  # shape: (n-(k-1)tau, k)

        # 3. Build epsilon grid, preferably by pairwise-distance quantiles
        D = pairwise_distances(Y)
        eps_grid = make_epsilon_grid(D, n_eps=n_eps)

        # 4. For each epsilon build VR complex and compute summaries
        complexes = []
        pc_values = []
        betti_values = []
        f_values = []

        for eps in eps_grid:
            K = vietoris_rips_complex(Y, eps, max_dim=max_dim)
            complexes.append(K)
            pc_values.append(point_complexity(K, ambient_dim=k))
            betti_values.append(betti_numbers(K, max_dim=max_dim))
            f_values.append(f_vector(K, max_dim=max_dim))

        # 5. Build Sigma matrix from category/point-complexity proxy
        Sigma = build_sigma_matrix(complexes, pc_values)

        # 6. Principal eigenvector and critical scale
        eigvals, eigvecs = np.linalg.eigh(Sigma)
        v = eigvecs[:, np.argmax(eigvals)]
        if np.sum(v) < 0:
            v = -v
        idx = np.argmax(v)
        eps_crit = eps_grid[idx]

        # 7. Store selected-scale features
        features_k = {
            "k": k,
            "eps_crit": eps_crit,
            "eps_index": idx,
            "point_complexity": pc_values[idx],
            "bettis": betti_values[idx],
            "f_vector": f_values[idx],
            "principal_vector": v,
        }
        all_features.append(features_k)

    # 8. Detect transition/stabilization over k
    p_hat = detect_stabilization(all_features)
    return p_hat, all_features
```

11.2 Recommended numerical safeguards

(1) Use a distance-quantile grid instead of a raw linear grid when comparing different embedding dimensions.

(2) Cap the maximum simplex dimension at 2 or 3 unless the sample size is small and

computation is manageable.

(3) Track computation time as k increases; Vietoris–Rips complexes can grow extremely quickly.

(4) Use sparse or approximate methods for large samples.

(5) Always repeat the analysis across multiple random seeds in simulation.

(6) For real data, use block bootstrap or rolling-window sensitivity analysis.

12 Interpretation of Possible Outcomes

12.1 Clear transition near p

If the feature increments are large for $k < p$ and small for $k \geq p$, the interpretation is straightforward: the embedding dimension p appears sufficient to capture the dependence structure. This is the ideal outcome.

12.2 Transition near $p + 1$

If the strongest transition appears at $p + 1$, this may be due to the recurrence constraint. The AR(p) equation relates X_t to p previous values, so the linear relation is visible in a cloud of $(p + 1)$ consecutive observations. In such cases one should report both candidates and explain the convention:

$$\text{state dimension } p \quad \text{versus} \quad \text{recurrence-detection dimension } p + 1. \tag{46}$$

A practical correction is to define $\hat{p} = k^* - 1$ if the chosen TDA feature is explicitly based on detecting the first recurrence constraint in $\mathbb{R}^k$.

12.3 No clear transition

If no clear transition appears, possible reasons include:

- sample size is too small,
- innovation variance is too large,
- the true process is not well approximated by an AR(p),
- the series is nonstationary,
- the ε grid is too coarse,
- high-dimensional distance concentration obscures the topology,
- the selected topological summaries are not sensitive to the AR order.

In this case, the correct conclusion is not that TDA failed, but that the data do not support a stable topological order estimate under the chosen configuration.

13 Connection with Your Scale-Selection Work

Your existing work uses synthetic point clouds such as circles and tori to study how Vietoris–Rips complexes evolve with ε, how LS-category-inspired quantities behave, and how a spectral Σ-matrix can identify a critical scale. The AR(p) problem keeps the same inner scale-selection problem but adds an outer model-selection problem:

$$\text{inner problem: choose } \varepsilon_{\text{crit}} \text{ for fixed } k, \tag{47}$$

$$\text{outer problem: choose } p \text{ by comparing dimensions } k. \tag{48}$$

Thus, the AR order estimation problem is naturally a two-level extension of the original TDA scale-selection pipeline.

13.1 Two-level objective

For every k, let

$$\mathcal{S}(k, \varepsilon) \tag{49}$$

be a scale-quality functional derived from the Σ-matrix, point complexity, Betti numbers, or a bilinear combination of summaries. The inner optimization is

$$\varepsilon_{\text{crit}}(k) = \arg\max_{\varepsilon \in \mathcal{E}_k} \mathcal{S}(k, \varepsilon). \tag{50}$$

The outer criterion is then based on the selected values

$$\mathcal{G}(k) = \mathcal{S}\big(k, \varepsilon_{\text{crit}}(k)\big) \tag{51}$$

or on the selected-scale feature vector $T(k)$. The order estimate is an elbow or stabilization point of $\mathcal{G}(k)$ or $T(k)$.

13.2 Using isolated vertices

In noisy point clouds, isolated vertices can represent outliers or rare states. In AR delay clouds, isolated points may correspond to large innovations, heavy-tailed shocks, regime changes, or boundary effects. Removing isolated vertices at $\varepsilon_{\text{crit}}(k)$ can make the topology more stable, but it should be reported carefully: for time series, extreme events may be statistically meaningful rather than mere measurement noise.

A good compromise is to compute both original and cleaned summaries:

$$T_{\text{raw}}(k), \qquad T_{\text{clean}}(k), \tag{52}$$

and estimate p from both. Agreement between raw and cleaned estimates increases confidence.

14 Limitations and Theoretical Cautions

14.1 No universal guarantee

The proposed TDA pipeline is a diagnostic and exploratory estimator. It does not, by itself, provide a universal consistency theorem for AR order estimation. A full theorem would need assumptions on stationarity, mixing, innovation tails, coefficient stability, growth of n relative to k, scale-grid construction, and stability of the selected topological functional.

14.2 Distance concentration

As k grows, Euclidean distances in $\mathbb{R}^k$ may concentrate. This can cause Vietoris–Rips complexes to change in ways unrelated to AR order. Normalizing scales by distance quantiles and limiting $k_{\max}$ are important safeguards.

14.3 Short time series

The number of embedded points is $N_k = n - (k-1)\tau$. As k grows, N_k decreases. Therefore, high embedding dimensions simultaneously increase ambient dimension and reduce sample size. This makes topology harder to estimate.

14.4 Stochastic versus deterministic dynamics

Takens' theorem is not a theorem about AR processes. It should be presented as motivation for delay-coordinate geometry, not as a formal justification of order recovery. The more direct justification for AR(p) is the Markov state representation in $\mathbb{R}^p$.

15 Recommended Reported Outputs

For a thesis or experimental section, report the following:

(1) The simulated or observed time series plot.

(2) PACF and AIC/BIC estimates as baselines.

(3) Delay-cloud plots for $k = 2$ and $k = 3$ when possible.

(4) For each k, the selected $\varepsilon_{\text{crit}}(k)$ or selected distance quantile.

(5) Betti curves or selected-scale Betti numbers across k.

(6) f-vector densities across k.

(7) Point-complexity curves across ε and selected values across k.

(8) The Σ-matrix heatmap for selected dimensions k.

(9) The principal eigenvector over the ε grid.

(10) The final transition plot used to choose $\hat{p}$.

16 Suggested Thesis Wording

A concise thesis paragraph could be:

> For each candidate embedding dimension k, the scalar time series was transformed into a delay-coordinate point cloud $\mathcal{Y}_k \subset \mathbb{R}^k$. A Vietoris–Rips filtration was then constructed over a grid of scales. To avoid choosing the scale manually, an LS-category-inspired similarity matrix $\Sigma^{(k)}$ was computed from the complexes in the filtration, and the critical scale $\varepsilon_{\text{crit}}(k)$ was selected from the dominant eigenvector of $\Sigma^{(k)}$. At this scale, topological and combinatorial descriptors such as Betti numbers, simplex counts, and point-complexity scores were recorded. The autoregressive order was estimated by identifying the first embedding dimension at which these descriptors stabilized. This procedure is motivated by the fact that an AR(p) process becomes Markovian in its p-lag state vector, so embeddings below p omit relevant state information, while embeddings above p mainly add redundant lag coordinates.

17 Conclusion

AR(p) processes have a natural state dimension p: the future depends on the previous p observations through a linear recurrence plus innovation. Delay embedding converts a scalar time series into a family of point clouds $\mathcal{Y}_k \subset \mathbb{R}^k$, allowing one to study how the geometry and topology of the reconstructed state space evolve with k. The TDA scale-selection pipeline based on Vietoris–Rips complexes, LS-category-inspired point complexity, a Σ-matrix, and the principal eigenvector gives a principled way to choose an informative scale for each embedded cloud. AR order estimation can then be formulated as a second-level problem: find the embedding dimension where selected-scale topological summaries stabilize or where a clear transition occurs.

The method is most defensible when used together with classical tools such as PACF, AIC, BIC, and prediction error. Its strength is not merely producing a single number $\hat{p}$, but providing a geometric diagnostic of memory, redundancy, noise, and possible model misspecification. Takens' theorem gives conceptual motivation for delay embeddings, but the direct AR justification comes from the p-dimensional Markov state representation. In practical work, the strongest evidence for a TDA-based order estimate is stability across scale grids, sample sizes, block resampling, and agreement with classical AR diagnostics.

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